

HAT-P-36b

R-0931

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230328 · created 2026-07-29T11:13:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460031.7007 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.027
Transit depth (Rp/Rs)^2	0.42 +/- 0.35 [%]
Orbital Inclination (inc)	89.54 +/- 2.72
Airmass coefficient 1 (a1)	1.045 +/- 0.019
Airmass coefficient 2 (a2)	-0.0308 +/- 0.0089
Scatter in the residuals of the lightcurve fit is	1.8 %
Best Comparison Star	#3 - [121, 299]
Optimal Aperture	4.42
Optimal Annulus	14.54
Transit Duration (day)	0.0906 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.065 $\pm$ 0.027 vs 0.126 (deviation 2.17 σ)
Duration fit vs published	0.0906 d vs 0.0925 d (deviation 0.52 σ)
Mid-time O-C	-4.7 min
Depth SNR	2.3
Residual scatter	1.8 %

Light curve

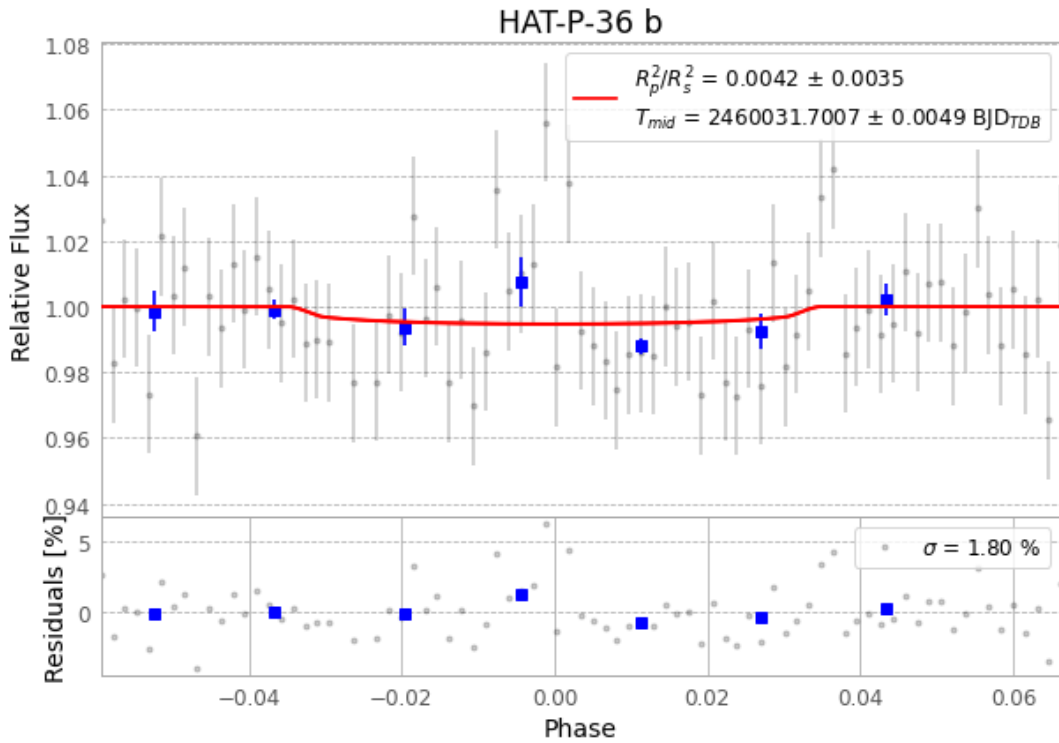


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-27.png (SHA-256 eeb98718e74a3755...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [552, 22], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 03e7d65736c8a03f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-36230328024815.FITS → HATP-36230328065115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230328.log (e9a84e6f451a...), FinalLightCurve_HAT-P-36 b_2023-03-27.png (eeb98718e74a...), FinalLightCurve_HAT-P-36 b_2023-03-27.csv (abe4eca2457b...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 11:13Z.